

MAOLEI HUANG

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👤 Profile

- **Business Experience:** Experienced in search and recommendation algorithms for a large-scale online travel platform. Lead the algorithmic framework for Fliggy's transportation discovery services, spanning flight and train search, cross-vertical transportation blending, and ancillary product selection and ranking.
- **Technical Expertise:** Long-term focus on personalized ranking, user intent understanding, sequential behavior modeling, and LLM-powered search and recommendation. Skilled at translating complex business constraints into learnable objectives, with end-to-end capabilities spanning problem abstraction, data analysis, scenario-specific modeling, online deployment, and experimental evaluation while balancing model effectiveness, latency, and system stability.

🎓 Education

University of Chinese Academy of Sciences (ICT, CAS)

Sep. 2019 – Jul. 2022

M.Eng. in Computer Architecture

Beijing, China

Shandong University

Sep. 2015 – Jul. 2019

B.Eng. in Software Engineering

Jinan, Shandong, China

👜 Professional Experience

Alibaba Group | Fliggy | Senior Algorithm Engineer
Transportation Discovery

Jun. 2021 (Intern) – Aug. 2022 (Full-time) – Present

- **Business Scope and Technical Ownership:** Lead Fliggy's search and recommendation algorithm framework for transportation discovery across four major areas: flights (search ranking, discounted-flight and nearby-flight recommendation), trains (search ranking), cross-vertical scenarios (transportation blending), and ancillary services (product selection and ranking). Own the full lifecycle from requirement analysis, problem formulation, and data construction to retrieval, ranking, online deployment, and experiment review, enabling unified capabilities and coordinated iteration across businesses and scenarios.
- **Framework Development and Scenario Abstraction:** Led the consolidation of algorithm systems and restructuring of serving pipelines, decoupling algorithm logic from engineering infrastructure and standardizing samples, features, and metrics across scenarios. Connected full-funnel behavioral signals including impressions, clicks, and payments, and established offline replay, hierarchical diagnosis, and scenario- and cohort-specific A/B testing. First proposed the LSTP framework for transportation behavior—Long Decision Cycle, Sparse Behavior, Temporal Patterns, and Periodicity—to characterize how travel differs from conventional e-commerce and guide sample construction, feature representation, model objectives, and iteration cadence.
- **Ranking Evolution and Scenario Modeling:** Progressively upgraded ranking from manually designed rules to GBDT/NN baselines and then to deep models centered on key decision factors such as price, journey duration, departure time, inventory, and capacity. Designed Similar Pairwise price gain, Monotonic Encoder context gain, U2G, Target Query Attention, and Listwise Ranking modules. Introduced contextual personalization, delayed-feedback correction, price-change awareness, and intent distillation to address nearby-flight context dependency, delayed conversion feedback, dynamic pricing, and ambiguous search intent. Under strict latency constraints, ensured effectiveness, performance, and stability through baseline fallbacks, staged rollout, and gradual replacement with deep models.
- **LLM-powered Search and Recommendation:** Explored end-to-end Generative Ranking that jointly models user profiles, long- and short-term behaviors, real-time queries, and candidate semantics; built an SFT + GRPO training pipeline with multi-objective rewards combining NDCG and Pay-NDCG to improve the understanding of ambiguous intent and complex itinerary comparisons. Also developed a transportation AI agent featuring ODD parsing, intent routing, multi-source retrieval, constraint-conflict resolution, and LLM reranking, completing the loop from natural-language understanding and cross-category planning to itinerary assembly and booking handoff, and advancing algorithms from Passive Ranking toward Proactive Planning.

Key Business Impact

- **Search Ranking:** Integrated inventory awareness, capacity stratification, price gain, multi-scenario routing, Listwise objectives, and DPP/MMR diversity reranking to comprehensively upgrade personalized flight and train ranking, delivering approximately +6% cumulative flight conversion and +4.5% cumulative train conversion.
- **Recommendation and Monetization:** Built the end-to-end architecture for discounted-flight recommendation and upgraded its models, increasing CTR by 20%. Developed SKU-level personalized ranking and dynamic pricing by device and user cohort for ancillary products, lifting ancillary revenue by 2.79% and generating tens of thousands of RMB in additional daily revenue across multiple business lines.
- **Research and Reuse:** Authored six first-author top-tier conference papers (four published: KDD'25 / AAAI'26 / SIGIR'26×2; two under review) and obtained two granted patents. The general ranking optimization framework, time-series forecasting framework, and LSTP modeling paradigm have been reused across multiple team scenarios.

Publications and Patents

Published Papers

- **CPNet**: A Context based Personalized Deep Network for Nearby Flight Recommendation.
“**Maolei Huang***, Detao Lv, Yao Yu, Shuhan Song, Dong Li, Zhuoran Zhuang” [**KDD’25, CCF-A**]
- **DCRNet**: Delayed Conversion Modeling based Personalized Flight Itinerary Ranking Network.
“**Maolei Huang***, Zhuoran Zhuang, Detao Lv, YT Li, Shuhan Song, Yao Yu” [**AAAI’26, CCF-A**]
- **SkyDistill**: Navigating Fuzzy Flight Search Ranking via Precise Intent Distillation.
“**Maolei Huang***, Huawei Cao, Shuhan Song” [**SIGIR’26, CCF-A**]
- **PCANet**: Price Change Aware Framework for Mitigating Inconsistencies in Large-Scale Ranking Systems.
“**Maolei Huang***, Huawei Cao, Shuhan Song” [**SIGIR’26, CCF-A**]
- **MTSSL**: Multitask Learning Using Feature Extraction Network for Smart Tourism Applications.
“Yu Li, Fanxiang Zeng, Nan Zhang, Zulong Chen, Li Zhou, **Maolei Huang**” [**IoT-J’23, SCI Q2**]
- **FMTL**: Multitask-Based Self-Supervised Learning for Recommendation in Social Systems.
“Wenjian Xu, Fanxiang Zeng, Nan Zhang, Yuyu Yin, Zl Chen, **Maolei Huang**” [**TCSS’24, SCI Q2**]

Papers Under Review

- **CRA**: A Tripartite Adaptation Framework for Robust Promotional Drift Mitigation in Industrial Recommendation. [Submitted to **KDD’27, Under Review**]
- **HiGAP**: Hierarchical Granularity Ambiguity-aware Personalized Recommendation. [Submitted to **AAAI’27, Under Review**]

Patents

- [**CN119669581B**] Method and Apparatus for Optimizing a Recommendation Model. [**First Inventor, Granted**]
- [**CN119669582B**] Ticket Recommendation Method, System, Travel Server, and Program. [**First Inventor, Granted**]

Technical Expertise

- **Search and Recommendation**: End-to-end expertise in retrieval, ranking, and reranking systems, with strengths in Learning to Rank, Pairwise/Listwise Ranking, multitask and multi-objective optimization, and unified modeling of user behavior, query intent, temporal feedback, and multiple scenarios.
- **LLM-powered Search and Recommendation**: Focused on Generative Ranking and AI agents, with hands-on experience in SFT, GRPO, multi-objective reward design, intent routing, multi-source retrieval, LLM reranking, and solution planning under complex constraints.
- **Production-grade Algorithm Deployment**: Experienced with large-scale training pipelines built on MaxCompute, Alibaba Cloud PAI, and distributed TensorFlow, as well as RTP/TPP online serving systems. Covers data and sample construction, offline diagnosis, A/B testing, staged rollout, latency optimization, and reliability assurance.